Daris Chen

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Education

University of California San Diego
Bachelor of Science (B.S.) in Computer Engineering — *San Diego, CA*

Work Experience

Lead Full-Stack Engineer — PatentIQ — Jan 2026 – Apr 2026
Product Manager Accelerator — *Remote*

- Orchestrated Agile development workflows via ClickUp for a 14 member cross functional team while serving as lead architect and primary contributor to deliver the foundational PatentIQ AI technical MVP.
- Constructed an AI-powered patent search and drafting assistant hosted on Vercel, EC2, and Supabase using Next.js, Auth0, and FastAPI to simplify intellectual property research for early-stage founders.
- Programmed a hybrid semantic search engine combining OpenAI text-embeddings and PostgreSQL pgvector to perform high-dimensional similarity ranking and recommendation system over global USPTO patent datasets.
- Built recursive SQL query builder with Zod-validated JSON filters and implemented an LRU cache to DB reduce latency.

Lead Full-Stack Engineer — Aug 2025 – Mar 2026
Groundwork Books — *San Diego, CA*

- Led architectural migration of production e-commerce to Next.js and Square for live inventory synchronization.
- Engineered a semantic search layer using Pinecone with integrated text embeddings, enabling topic and concept-based discovery across 4,000+ SKUs without requiring exact title matches.
- Architected a two-tier caching system: a look-aside Redis reducing Square API latency by 90% (sub-200ms p99), and a client-side request coalescing hook batching 50+ concurrent inventory lookups into an API call with 2-minute TTL cache.

FishSense Artificial Intelligence Researcher — Feb 2025 – Mar 2026
UCSD Engineers for Exploration (E4E) — *San Diego, CA*

- Engineered a custom R-CNN training pipeline (ResNet-50 backbone) for underwater laser-profiling, standardizing the Detectron2 codebase, to reducing inference setup time by 40% for new researchers and increasing reproducibility.
- Developed a dual-laser photogrammetry algorithm to extract biological metrics (fish length/mass) from 2D video feeds, achieving sub-5% error rates in pixel-to-centimeter conversion.

Software Engineering Intern — Jan 2025 – Sep 2025
Personify AI (Pre-Seed EdTech) — personifyapp.ai — *Remote*

- Refactored feature leaderboard user interface using Next.js, Tailwind CSS, and Firebase, increasing user traffic by 15%; designed duplicate-detection logic in Firebase to clean the data pipeline.
- Built admin controls for approvals, rejections, and closures, and added vote-based ranking, category sorting, and search, cutting resolution time by 50% and reducing user interaction time by 20%.
- Engineered a data ingestion pipeline for unstructured text, implementing deduplication logic and NLP-based feature extraction to structure dataset inputs for the recommendation engine.

Projects

FlipperZillow | *Next.js, Python, PyTorch, AMD ROCm, Three.js, WebSpatial* — Feb 2026 – Mar 2026

- Constructed an AI house tour platform named FlipperZillow using Next.js and Claude for semantic search, piping DFormerV2 spatial analysis into Claude Sonnet 4.6 to generate property scripts and ElevenLabs audio narration.
- Integrated an AMD ROCm SSH pipeline to process Realtor.com property images cascading DepthAnythingV2 into DFormerV2 and extracting WebSpatial compatible 3D objects using META's SAM3D Object.
- Assembled a 3D experience combining Google Maps and WebSpatial for Vision Pro with a Three.js fallback.

Mini-Stockfish Chess Engine | *Pygame GUI, Torchscript, NumPy, Syzygy/Gaviota* — Mar 2025 – Mar 2026

- Architected a hybrid chess engine combining HalfKP NNUE neural network with advanced alpha-beta search featuring killer moves, counter-move heuristics, history heuristics, quiescence search, NMP, and LMR.
- Optimized critical path using Cython, C++ legal move generation, and Zobrist hashing, reducing leaf node evaluation latency by 77.2%, cutting eval time from 1.1 ms to 0.25 ms on single-threaded commodity hardware.

RedShift LLM Jailbreak | *Python, PyTorch, CUDA, Prompt Engineering* — Jan 2025 – Mar 2025

- Engineered an automated adversarial attack framework ("Red Teaming") for LLMs, implementing Chain-of-Thought injection techniques to bypass safety filters on models (ChatGPT, Llama, Vicuna, DeBERTa, DeepSeek, Grok).
- Built a scalable evaluation pipeline using PyTorch DataLoaders to batch-process adversarial prompts, increasing attack coverage by 67% across 7 major model architectures.

Technical Skills

Programming Languages: Python, Java, C/C++, JavaScript/TypeScript, HTML/CSS, ARM Assembly, SQL, Verilog/SystemVerilog, Go, Cython
Libraries & Frameworks: PyTorch, TensorFlow, CUDA, ROCm, cuDNN, Transformers, NumPy, pandas, Matplotlib, Scikit-learn, Detectron2, tqdm, React, Next.js, Node.js, Express.js, Three.js, Svelte, Tailwind CSS, Zod, DepthAnythingV2, DFormerV2, SAM3D Object, OpenAI, Claude, Signal Processing, FFT
Tools & Technologies: Git, GitHub Actions, Docker, Kubernetes, Linux, Clickup, Jira, Jest, Puppeteer, GCP, gRPC, EC2, GCC, OpenCL, Figma, VS Code, CodeMirror, Wandb, Jupyter, Hugging Face, Vercel, ESLint, Turbopack, OSM Nominatim, Open-Meteo, OpenStreetMap, Intel FPGA ModelSim, WebSpatial (Vision Pro), Auth0, SSH, HB100, RF Engineering, ESP32
Databases: PostgreSQL, MongoDB, Upstash Redis, Firebase, Pinecone, pgvector, Supabase